

■ 2.8 Advanced Topic: Resource Management

■ 2.8.1 Time

`Timing[expr]` evaluate *expr*, and return a list of the result, and the CPU time it took

Measuring computer time.

The actual time you wait to get the result of a calculation in *Mathematica* is a combination of several components.

First, there is the actual CPU time that is spent doing the computation. This is the time that is measured by `Timing`.

Second, there is time that your operating system spends in swapping *Mathematica* code and data on and off your mass storage device. On many computer systems, you will notice an extra delay the *first* time you use a particular *Mathematica* function. The extra time is spent in reading the code for the function off the mass storage device.

If your computer has a multi-tasking operating system (such as UNIX), the total time that elapses during a *Mathematica* computation also depends on the other things that your computer is doing.

One thing you should realize is that even the CPU time spent on a particular computation may be different the second time you do a computation. Many built-in *Mathematica* functions, for example, build internal tables only when you first use them in a particular way. Once these tables have been built, the functions will work much more quickly.

There are also a number of crucial optimizations that *Mathematica* uses internally to avoid re-evaluating expressions when it is not necessary. If you do the same calculation several times, it is possible that certain re-evaluations will be done on some, but not all, occasions. The result is that the CPU time spent doing the computation may vary.

This gives the CPU time used in computing 1000! on a particular kind of computer. The `;` stops the value of the factorial from being printed, and returns `Null` instead.

```
In[1]:= Timing[1000!;]
Out[1]= {1.73333 Second, Null}
```

When you first ask for the factorial, *Mathematica* builds some internal tables. The second time you ask for it, it is much faster.

```
In[2]:= Timing[1000!;]
Out[2]= {0. Second, Null}
```

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